

- Concise Functional Requirements Document (FRD) for the "Application"***

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Overview

The "Application" is a data processing pipeline designed to load customer and order data from CSV files into Delta tables, perform data cleansing, and create aggregated summaries.

Functional Requirements

1. **Data Ingestion**

- Read CSV data from specified locations and load it into Delta tables "customer" and "order".
- CSV file paths: `\\Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and `\\Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`.

2. **Data Cleansing**

- Remove "Null"/null and duplicate records from "customer" and "order" tables.

3. **Data Transformation and Loading**

- Create "ordersummary" table if it does not exist in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- Join "customer" and "order" data using "CustId" field and load into SCD type 2 table "ordersummary".

4. **SCD Type 2 Implementation**

- Update SCD type 2 "ordersummary" table when there are changes in "customer" table.
- Make old records inactive and new records active.
- Update `StartDate` and `EndDate` accordingly.

5. **Aggregated Summary**

- Create "customeraggregatespend" table if it does not exist in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- Aggregate "TotalAmount" from "ordersummary" table and group by "Name" and "Date" columns.
- Load aggregated data into "customeraggregatespend" table.

Diagrams

System Flow Diagram

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Process Flow / Flowchart

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Data Flow Diagram (DFD)

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Sequence Diagram

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High-Level Architecture Diagram

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